

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	12 July 2026
Team ID	SWTID-2026-6951
Project Name	Visualization tool for electric vehicle charge and range analysis
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import EV datasets (CSV files) into MySQL and connect them to Tableau. Validate imported data before visualization.
FR-2	Dashboard Visualization	Display interactive dashboard with EV analysis. Show charts for Top Speed, Body Style, Efficiency, Brands, and Powertrain Type.
FR-3	Story Presentation	Display Tableau Story with step-by-step insights. Include navigation between Dashboard and Story.
FR-4	Data Filtering	Allow users to filter data by Brand, Body Style, Powertrain Type, Region, and Charging Station Type. Update charts dynamically based on selected filters.
FR-5	Web Application	Embed Tableau Dashboard and Story into Bootstrap website. Provide responsive interface for users.
FR-6	Data Analysis	Compare EV brands, charging stations, efficiency, prices, and top speed. Present insights through interactive visualizations.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR1	Usability	The application should be easy to navigate with a simple and user-friendly interface.
NFR2	Security	The dashboard should use published Tableau Public links without modifying the original data.
NFR3	Reliability	The dashboard should display accurate and consistent visualizations based on the uploaded datasets.
NFR4	Performance	Dashboard and Story should load quickly and respond smoothly to filters.

